Tarek Messaoudi

Full Stack Engineer

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Education

IMT Lille-Douai **2012 - 2017**

Software engineer major // with a foreign semester in South Korea at Chugnam National Univ.
Internships at Capgemini, GFI, Alliance Française Limassol, Lille 2 University, Norauto

Technical Skills

- **Programming Languages:** TypeScript, JavaScript, Python, Java
- **Frontend Development:** React, Tailwind CSS, Remix
- **Backend Development :** NodeJs, Deno, FastApi, Spring
- **Databases :** PostgresSQL, SQLite, MongoDB, REDIS
- **Other Technologies:** Docker, Kubernetes, Git, Linux

Work Experience

Software Engineer **2026 - Present**

Chipiron // An MRI scanner company

🖥️ MRI Operator Web Application

- Built the app from the first commit as its main developer, shipping 14 feature releases in six months
- Designed the exam workflow: drag-and-drop protocols, step-based execution, live scan progress, abort
- Built a JSON Schema-driven form engine for acquisition parameters, with server-side field validation

🫁 DICOM Imaging Integration

- Connected Orthanc PACS storage, an authenticated DICOMweb proxy and a Cornerstone.js viewer

🔐 Access Control & Quality

- Implemented authentication and role-based access control in a FastAPI service
- Introduced front-end testing from scratch (Vitest, MSW, Playwright) and OpenAPI-generated API types

Full Stack Engineer **2020 - 2025**

Qynapse // A healthcare technology company specializing in neuroimaging and AI

🧠 Medical Imaging Web Platform Redesign

- Refactored with a new admin panel, interactive data visualizations, and a robust MRI viewer, ...
- Added features such as segmentation correction, lesion tracking, 3D rendering, oblique views, multi-modality
- Maintained long-term stability of the viewer through refactors and testing

📝 Reporting System Overhaul

- Rebuilt the report generation pipeline using Jinja2 + Puppeteer (HTML → PDF)
- Redesigned templates for multiple languages and disease-specific variations (e.g., Dementia, MS, ...)

📦 Scalable Infrastructure & Microservices

- Contributed to converting the academic BrainVisa platform into a Kubernetes-based scalable pipeline
- Maintained and expanded the customer-deployed DIMSE agent, adding routing logic and detailed logging

🧬 Domain-Specific Tools

- Built a normative Python library to compute statistical imaging metrics (z-scores, percentiles, asymmetry, ...)
- Developed a 3D mesh microservice for structural MRI data

Full Stack Engineer **2017 - 2020**

Alter Solutions // A consulting and IT services company

🔬 *IRT System X : Central Dashboard promoting research project webservices*

- Develop the whole application from scratch

☎️ SFR Altice Group: Brand-new customer service web app

- Refactor the whole custom CRM with new technologies and architecture
- Added core features (Mail/SMS notification, Tray system, ...)

🪖 Thalès: Local web app displaying military data from Excel file

- Create a user-friendly tool for data vizualizing inventory file (.xlsx)

🚙 PSA: Online tool to parse automobile file format (.a2l and .hex, .ulp)

- Set up a backend service for parsing efficiently raw data files

Projects & Rewards

LOM (GitHub) - *Browser strategy game*

"Entreprise du Futur" Reward (Site) - *By Thalès, from Challenge Entreprendre*